

Orthogonality

EE263
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Orthonormal set of vectors

set of vectors $\{u_1, \dots, u_k\} \subset \mathbb{R}^n$ is

- ▶ *normalized* if $\|u_i\| = 1$, $i = 1, \dots, k$
(u_i are called *unit vectors* or *direction vectors*)
- ▶ *orthogonal* if $u_i \perp u_j$ for $i \neq j$
- ▶ *orthonormal* if both

slang: we say ' $u_1, \dots, u_k$ are orthonormal vectors' but orthonormality (like independence) is a property of a *set* of vectors, not vectors individually

in terms of $U = [u_1 \ \cdots \ u_k]$, orthonormal means

$$U^T U = I_k$$

Orthonormality

an orthonormal set of vectors is independent

- ▶ to see this, multiply $Ux = 0$ by U^T
- ▶ hence $\{u_1, \dots, u_k\}$ is an *orthonormal basis* for

$$\text{span}(u_1, \dots, u_k) = \text{range}(U)$$

- ▶ **warning:** if $k < n$ then $UU^T \neq I$ (since its rank is at most k)
(more on this matrix later ...)

Orthonormal basis for $\mathbb{R}^n$

A matrix U is called *orthogonal* if

$$U \text{ is square and } U^T U = I$$

- ▶ the set of columns $u_1, \dots, u_n$ is an orthonormal *basis* for $\mathbb{R}^n$
- ▶ (you'd think such matrices would be called *orthonormal*, not *orthogonal*)
- ▶ it follows that $U^{-1} = U^T$, and hence also $UU^T = I$, i.e.,

$$\sum_{i=1}^n u_i u_i^T = I$$

Expansion in orthonormal basis

suppose U is orthogonal, so $x = UU^T x$, i.e.,

$$x = \sum_{i=1}^n (u_i^T x) u_i$$

- ▶ $u_i^T x$ is called the *component* of x in the direction u_i
- ▶ $a = U^T x$ *resolves* x into the vector of its u_i components
- ▶ $x = Ua$ *reconstitutes* x from its u_i components
- ▶ $x = Ua = \sum_{i=1}^n a_i u_i$ is called the (u_i-) *expansion* of x

Complementary subspaces

if $Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}$ and Q is orthogonal then $\text{range}(Q_1)$ and $\text{range}(Q_2)$ are called *complementary subspaces*, because

$$\text{range}(Q_2) = \text{range}(Q_1)^\perp$$

- ▶ they are orthogonal *i.e.*, every vector in the first subspace is orthogonal to every vector in the second subspace
- ▶ every vector in $\mathbb{R}^m$ can be expressed as a sum of two vectors, one from each subspace
- ▶ each subspace is the *orthogonal complement* of the other

Complementary subspaces

$$\text{range}(Q_2) = \text{range}(Q_1)^\perp$$

- ▶ $\text{range } Q_2 \subset (\text{range } Q_1)^\perp$ because $Q_1^\top Q_2 = 0$
- ▶ to show $\text{range } Q_2 \supset (\text{range } Q_1)^\perp$, suppose $x \in (\text{range } Q_1)^\perp$, then $Q_1^\top x = 0$, and since $x = Q_1 Q_1^\top x + Q_2 Q_2^\top x$ we have $x = Q_2 Q_2^\top x$ and so $x \in \text{range } Q_2$

Geometric interpretation

if U has orthonormal columns then transformation $w = Uz$

- ▶ preserves *norm* of vectors, *i.e.*, $\|Uz\| = \|z\|$
- ▶ preserves *angles* between vectors, *i.e.*, $\angle(Uz, U\tilde{z}) = \angle(z, \tilde{z})$
- ▶ we say U is *isometric*, it preserves distances